Selective Deflation, Not Displacement: Characterizing the Alignment Tax Under Length-Normalized DPO

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Abstract

Length-normalized DPO with an SFT retention term drives the implicit reward margin to $\sim$7 β-scaled units in a 4B+LoRA model trained on structured exact-format tasks. The dominant framing—displacement theory, IPO's design motivation—predicts this should degrade capability uniformly through chosen-likelihood decline. In a pre-registered 8-seed study with artifact-aware scoring, we find the alignment tax is real but graded: ranging from undetectable (extraction, $\Delta = -0.5$, TOST equivalence PASS) through small but statistically genuine (json, $\Delta = -3.5$, one-directional loss, TOST boundary FAIL) to catastrophic (router, $\Delta = -30$), tracking format-alignment to the preference data more than margin magnitude (training backend nf4; faithful-precision fp16 re-measurement yields margins of $\sim$3). The retention term anchors chosen-completion likelihood while DPO gradients suppress rejected completions; the entire margin is rejected collapse, channeled predominantly through the preference data's style axis (aggregate dominance under both backends; per-pair ordering is backend-dependent, §4.3). Gold-logp deflation is task-selective (extraction -0.15, json -1.04, router -1.47 nats/token, fp16), directionally consistent with behavioral outcomes, with gold selectively protected relative to incorrect controls on extraction (8/8 seeds). Artifact-aware scoring and pre-registered TOST equivalence testing prevented a false-collapse conclusion from a naive scorer. Scope: LN rewards with retention, sparse LoRA on 8/32 layers, 4B model, easily-separable synthetic preferences.

1 Introduction

DPO's implicit reward margin grows large under length-normalized rewards—to $\sim$7 β-scaled units in our experiments—echoing the unbounded growth IPO was designed to prevent. The dominant framing treats this as pathological: Razin et al. [11] identify likelihood displacement, where chosen log-prob decreases and probability mass may flow to out-of-distribution completions; IPO [1] bounds the margin, treating unbounded growth as the core problem; and Cal-DPO [13] observes absolute likelihood deflation and proposes a calibration fix. Yet the empirical basis comes primarily from judge-based evaluation or full fine-tuning at large scale. The common practical regime—small model, LoRA, length-normalized rewards, structured exact-format tasks—is understudied. A methodological gap compounds the empirical one: the literature lacks equivalence testing for capability preservation, sometimes reporting "no significant difference" without the power to distinguish "preserved" from "underpowered to detect a real drop."

We present a pre-registered 8-seed study measuring behavioral capability (free-generation with artifact-aware scoring) and probability-layer dynamics (per-token gold/control log-probabilities) as the margin grows to $\sim$7 (nf4 training backend; $\sim$3 under fp16 re-measurement) in a 4B model fine-tuned with LoRA. Our objective combines length-normalized DPO with an SFT retention term (weight 0.5) that anchors chosen-completion likelihood—a common practical choice that, as we show, channels DPO's optimization pressure entirely into rejected-completion suppression.

The alignment tax is real but graded: undetectable on extraction ($\Delta = -0.5$, TOST equivalence PASS), small but statistically genuine on json ($\Delta = -3.5$, one-directional loss), and catastrophic

on router ($\Delta = -30$). The gradient tracks format-alignment to the preference data's dominant style axis, consistently across all 8 seeds. With three tasks, we report this co-occurrence without claiming causation.

The mechanism is asymmetric: the retention term keeps chosen reward approximately stationary while DPO gradients drive rejected reward to -14.2 (nf4), producing margin growth entirely through rejected-completion collapse. The style axis dominates this suppression in aggregate under both backends; per-pair ordering is backend-dependent (§4.3)—the pre-registered prediction that style would dominate held-out per-pair margin at every checkpoint was falsified under both backends (fp16 0/4; nf4 2/4). Gold-logp deflation is itself task-selective and directionally consistent with behavioral outcomes, though non-linearly related (§4.3–4.4). A naive strict-match scorer masked this picture, producing a false capability collapse ($0.616 \rightarrow 0.988$ under canonicalized scoring) that artifact-aware evaluation and pre-registered TOST equivalence testing—used here as the primary evaluation framework for capability assessment—prevented.

Our contributions are: (1) a graded, task-selective alignment tax driven by rejected-collapse dynamics and style-axis aggregate pressure, with non-monotonic format-break dynamics at margin ≈ 3 and a falsified pre-registered prediction on held-out per-pair margin (§4.1–4.3); (2) task-selective gold-logp deflation with selective protection demonstrated on extraction (8/8 seeds, paired statistic), robust across quantization backends (§4.4); and (3) an artifact-aware evaluation protocol with pre-registered TOST equivalence testing, whose falsification chronology (three falsified hypotheses, two refined characterizations) documents how pre-registration forced successive refinement of initially simpler claims (§3, §5.2).

This study operates within explicit scope conditions: sparse LoRA on 8/32 layers (0.034% of parameters; not full fine-tuning), a single 4B model, length-normalized rewards with an SFT retention term (weight 0.5), easily-separable synthetic preferences, and three structured exact-format tasks. We do not study safety or refusal tasks—where Razin et al. demonstrated catastrophic displacement—full fine-tuning, naturalistic human-written preferences, or models above 4B parameters. These are stated boundaries, not implicit weaknesses.

2 Related Work

Likelihood displacement. Razin et al. (ICLR 2025) [11] identify likelihood displacement under DPO—chosen log-prob decreases while mass may flow to out-of-distribution or harmful completions—and introduce the CHES score to flag pairs whose embedding proximity drives catastrophic displacement, which their analysis notes is "not necessarily problematic" (catastrophic only when mass flows to opposite-meaning responses). Our SFT retention term (equivalent to RPO; Liu et al. [6]) anchors chosen likelihood—the mitigation Razin et al. themselves propose—so this chosen-decline dynamic is largely suppressed (§4.3); whether displacement would emerge without it is untested. Pipano et al. [8] characterize which f-divergence losses are displacement-resistant by construction; our approach instead retains the standard DPO loss and relies on the SFT term for anchoring.

The squeezing effect. Ren and Sutherland (ICLR 2025) [12] show DPO's gradient is dominated by negative gradients on rejected completions, concentrating mass onto the highest-probability token per position. They demonstrate this at the single-token level on Pythia and Qwen; we observe consistent dynamics on multi-token structured completions, though the transfer is assumed rather than proven. Our reward decomposition (§4.3) is textbook squeezing (rejected -14.2, chosen within -0.072 to $+0.052$, nf4), and offers a candidate explanation for task-selective gold deflation (§4.4): extraction outputs share entities with training data and may gain from mass concentration onto gold tokens, while router's terse format, maximally dissimilar from the verbose pairs, may not.

Ordering preservation and calibration. Cal-DPO (Xiao et al., NeurIPS 2024) [13] observes DPO preserves preference ordering while absolute likelihoods deflate, and calibrates the implicit reward to prevent chosen-reward decline. Our retention term achieves similar anchoring (§3.1); chosen-flatness does not demonstrate Cal-DPO's concern is unfounded but that it is already addressed—and even with chosen anchored, the behavioral tax remains on off-target tasks (§4.1), driven entirely by rejected suppression (§4.3).

Bounding the margin. IPO (Azar et al., AISTATS 2024) [1] replaces DPO's log-sigmoid with a squared loss to bound the margin, treating unbounded growth as the core pathology. Our margin

grows to $\sim$7 with graded, format-selective effects rather than the uniform catastrophe bounding targets (§4.1–4.2); this does not invalidate IPO but characterizes a regime where damage concentrates in a margin 1–3 window and is task-selective.

Chosen-reward predictions. Rafailov et al. (COLM 2024) [10] predict from a Q-function framework that chosen log-prob should decrease under DPO; our retention term largely counteracts this at the endpoint, though faithful-precision measurement reveals a residual transient dip (§5.3).

Equivalence testing and scoring. Manzoni (2026) [7] uses TOST with 8 seeds for SFT capability preservation in MoE fine-tuning, and a concurrent paper (Bao and Abisado, *Symmetry* 2026) [2] applies TOST as incidental validation in a DPO context; to our knowledge, pre-registered TOST as the primary evaluation framework for DPO capability assessment is unprecedented in the DPO literature, though well-established in bioequivalence. Baxi (2025) [3] separates constraint compliance from semantic accuracy, establishing the decomposition; our contribution is the trajectory dynamics (transient format-break spike vs monotonic content-error rise, §4.2). Minerva (Lewkowycz et al., NeurIPS 2022) [4] precedents answer-normalization preventing exact-match false negatives; our scorer artifact (0.550 $\rightarrow$ 0.981, seed0) is a concrete alignment-specific instance.

DPO dynamics and alignment-tax context. Li (2026) [5] finds none of 20 DPO variants outperforms vanilla DPO under Bonferroni correction. Young (2026) [14] frames the alignment tax as the projection of the safety direction onto the capability subspace—our format-axis-alignment finding (§4.1) is an empirical instantiation. Pollanen (2026) [9] reports margin can anticorrelate with reasoning capability ($r = -0.91$, LLaMA on logic tasks, sweeping β); they vary β rather than observing within-run growth, making it related but distinct from our trajectory analysis.

3 Method

3.1 Setup

We fine-tune Qwen3.5-4B with LoRA ($r = 8$, $\alpha = 16$, dropout $= 0.05$) using vanilla length-normalized DPO ($\beta = 0.5$) with an SFT retention term. Qwen3.5-4B is a hybrid architecture with 24 linear-attention layers and 8 full-attention layers (every 4th: layers 3, 7, 11, 15, 19, 23, 27, 31); LoRA adapts only the 8 full-attention layers (q/k/v/o_proj; 1.57M trainable parameters, 0.034% of 4.66B total), leaving the 24 linear-attention layers and all MLPs untouched. The total loss is $L = L_{\text{pref}} + 0.5\,L_{\text{sft}}$, where L_{sft} is an assistant-only cross-entropy on the chosen completion (system and user tokens masked). The retention term anchors chosen-completion likelihood during preference optimization. This formulation corresponds to Regularized Preference Optimization (RPO; Liu et al. [6]), which adds an SFT loss on chosen responses to the DPO objective; Razin et al. [11] identify this SFT regularization as a mitigation for likelihood displacement. Training uses AdamW (lr 5×10^{-5}, constant schedule, weight decay 0.01), gradient clipping 1.0, batch size 8, and 180 fixed optimizer steps with no loss-based early stop and a divergence guard (abort on NaN or smoothed margin > 20). Eight seeds (0–7) start from the same SFT checkpoint (hash-verified; tokenizer and chat template byte-identical; Appendix E).

The preference data is 647 pairs (549 train, 98 held-out; deterministic split) on two axes: a correctness axis (200 tool_calling pairs, 171 in the 549-pair training split, where rejected completions are wrong-tool swaps—identical arguments, randomly substituted tool name, length-matched at $\sim$7.5 words) and a style axis (447 pairs, 378 in the training split, where chosen completions are verbose analytical responses ($\sim$190–226 words) and rejected are GPT-4o-mini degradations with numbers, dates, and section references stripped and text truncated to $\sim$86–100 words). Length normalization is applied because otherwise the 447 style pairs would be winnable on length alone. Seeds 0–4 ran in the first quota window and 5–7 in the second, pre-registered and inspected only after all 8 completed.

3.2 Tasks and Evaluation Suite

We evaluate three structured exact-format tasks, 160 items each, on SEC-filing queries (below). Free generation uses greedy decoding with no constrained decoding and no LLM judge; all behavioral generations are produced under nf4 quantization, matching the training backend, so the TOST verdicts describe the deployed-quantization regime end-to-end. max_new_tokens = 16 (router) or 128 (json, extraction). The 16-token router limit truncates a few FORMAT_BREAK outputs (6/251 at margin3) but does not affect FORMAT_BREAK classification, which depends on the leading token.

The baseline (baseline_v2) is the step-0 SFT checkpoint on the same suite and scorer, evaluated deterministically.

Task	Output format	Example	Pref. coverage
Json	Structured JSON tool call (function name + typed args)	`{"name": "get_filing", "args":{"ticker": "AAPL"}}`	On-target (200 tool_calling pairs)
Router	Bare UPPER_SNAKE_CASE label from a fixed 5-label set (FILING_ANALYSIS, FILING_DIFF, COMPANY_DEEP_DIVE, SIMPLE_QUERY, OUT_OF_SCOPE)	`COMPANY_DEEP_DIVE`	Off-target (zero)
Extraction	Three-field JSON: ticker, section, filing_date	`{"ticker":"MSFT", "section":"Item 7", "filing_date": "2024-02-15"}`	Off-target (zero)

Scoring uses scorer v2, a frozen three-tier scheme: PASS (semantically correct), SURFACE_FORM (correct content, surface-level format difference such as "Item 1" rendered as "1"), and SEMANTIC_ERROR (wrong content); FORMAT_BREAK (unparseable) is tracked separately. The primary metric is semantic accuracy (PASS + SURFACE_FORM). The extraction canonicalization rule—section-identifier matching via regex `\d+[A-Z]?`, case-folded, ignoring appended titles—was frozen before the matrix (Appendix B).

Scorer artifact (§3.2)

Under strict string matching, extraction accuracy at margin3 appears to crash, but canonicalized scoring recovers it: seed0 $0.550 \rightarrow 0.981$ ($69/72 = 96\%$); 8-seed mean $0.616 \rightarrow 0.988$. The flipped items are surface-form register shifts ("Item 1" vs "1"), not semantic failures—the difference between "catastrophic collapse at margin $\approx$3" and "capability preserved with a transient register shift."

3.3 Suite Independence and Domain Overlap

No eval prompt appears verbatim in the preference or SFT data (exact-string match), but both derive from EDGAR ticker$\times$section fetches, producing domain overlap by design. We report the most specific overlap available: extraction shares a ticker$\times$section combination with a preference pair's source filing for $\geq$81/160 items (ticker$\times$section provenance recoverable for 209/647 pairs); json shares a ticker for $\geq$131/160 items (section is not a json field). Extraction's preservation may therefore partly reflect entity familiarity rather than format-robustness alone; router, with zero domain overlap, provides the cleanest independence test. Degeneracy floors (extraction 0.006, router 0.200, json 0.0125), class balance, zero duplicate prompts, and gold self-consistency (160/160 per task) are reported in Appendix D.

3.4 Pre-Registered Analysis Plan

The primary endpoint is per-task semantic accuracy at the final checkpoint (step 180) versus baseline_v2, tested via TOST equivalence at $\pm$5 points. We chose $\pm$5 as an a priori bound large enough to avoid trivially passing ($n = 160$ items $\times$ 8 seeds gives reasonable power to detect 5-point differences) yet small enough that passing is substantively meaningful (a 5-point drop corresponds to 8 items flipping, which we judged non-negligible). The bound was frozen before any matrix data existed. We acknowledge this is researcher judgment, not derived from an external standard.

Confidence intervals are constructed at two levels: seed-level (mean per-seed difference, t-based CI, df $= 7$) and item-level (bootstrap over items, seeds pooled). Both must lie within $\pm$5, and $\geq$7 of 8 seeds must individually fall within $\pm$5, for a task to pass. If all three tasks pass, the bounded-null headline holds; if any fails, the task-selective branch governs. No multiple-comparison correction

is applied: the headline requires all tasks to pass (an intersection-union test), which is inherently conservative.

Final is saved at the fixed step 180; margin1/3/5 are saved when the 20-sample smoothed training margin first crosses 1.0/3.0/5.0 (steps vary by seed: margin1 24–30, margin3 84–98, margin5 112–138), with the smoothed margin at the save step reported as each checkpoint's margin label. Cross-seed averages at these checkpoints pool seeds at matched margin but different optimizer steps; the primary endpoint (step 180) and step20 diagnostic are unaffected. Step20 is excluded from all endpoints. All outcome-to-interpretation mappings were frozen and date-stamped before the decisive runs (Notes 1–2, committed 2026-06-11 and 2026-06-18).

3.5 Probability-Layer Measurement

This study uses two quantization backends. *nf4* is the training and behavioral-evaluation backend: all training, checkpoint saving, behavioral generation (§4.1–4.2), and the reward decomposition (§4.3) were performed under nf4. *fp16* is the faithful-precision audit backend: gold-logp measurement (§4.4), the GOLD_PROTECTED paired statistic, KL trajectory, and the held-out margin recomputation were performed under fp16 on Mac. Cross-backend magnitude comparisons are not valid; within-backend comparisons are used throughout; cross-backend ratios appear only in the §4.5 sensitivity analysis.

To assess whether training selectively protects gold completions relative to incorrect alternatives, we teacher-force three completion types through each checkpoint: gold (correct), control A (well-formed, different section identifier), and control B (well-formed, different ticker)—both structurally valid but semantically wrong. The primary statistic is the paired difference $\Delta_{\text{gold}} - \Delta_{\text{mean}(A,B)}$, where Δ is the change in per-token log-probability from the SFT baseline; a positive stat indicates gold deflates less than controls (selective protection). The pre-registered threshold is $X = 0.05$ nats/token, with the 95% bootstrap CI required to exclude zero. Controls A and B are extraction-specific; on json and router, gold-logp is reported as absolute Δ_{gold} without a paired baseline. A full reconciliation of all pre-registered secondaries against their status is in Appendix C.

4 Results

All 8 seeds completed 180 fixed optimizer steps without triggering the divergence guard. Final margins range 6.08–7.68 (mean 6.73) under the nf4 training backend ($\beta = 0.5$; checkpoint margin labels are training-time smoothed values; Table 3 reports the reward decomposition over all 549 train pairs, which yields slightly different values, e.g. 6.73 smoothed vs 7.12 decomposed at final). Within each backend, held-out preference margin tracks training margin closely (nf4: 7.12 vs 6.98, within 2%; fp16: 2.95 vs 2.78, within 6%; Appendix F), ruling out train-set memorization as the source of margin growth. Absolute margins differ 2.4× (train) to 2.5× (held-out) between backends, up to 3.4× on the style axis—a quantization sensitivity analyzed in §4.5.

4.1 Graded, Task-Selective Degradation

Table 1 reports the primary TOST verdicts. The pre-registered falsifier fires: router degrades far beyond ± 5, the bounded-null headline is dropped, and the task-selective branch governs.

Extraction passes all three criteria comfortably (point estimate -0.55, both CIs well within ± 5, 8/8 seeds inside the bound); a per-item flip analysis finds no systematic asymmetry (15 gains vs 22 losses, sign test, $p = 0.32$). Capability is preserved as the margin grows to ~ 7. Json fails, though marginally and ceiling-driven: the point estimate (-3.52) and seed-level CI are within ± 5, but the equivalence test fails because the item-level pooled CI lower bound (-5.31) crosses -5—not because of the seed count, even though only 7/8 seeds fall inside (seed0 at -6.25 outside; seed4 at -5.00, at the inclusive boundary, passes). The per-item flips are 0 gains and 45 losses (McNemar with continuity correction, $p = 5.4 \times 10^{-11}$)—every flip a regression, never offset by a gain. The degradation is statistically genuine, one-directional, and consistent, but small in absolute magnitude; the TOST failure is partly an artifact of testing near the ceiling (baseline 98.75), but the drop is real. Router degrades catastrophically (-30.16, both CIs far outside ± 5, 0/8 seeds inside); the endpoint distribution is roughly symmetric (mean 0.636, median 0.638, range 0.488–0.769) but wide (spread 0.281), reflecting high seed-to-seed variance despite consistent damage depth (§4.2).

Table 1: Primary TOST equivalence test (±5 pts, semantic accuracy at step 180 vs baseline_v2).

Task	Base	Final	Δ	Seed 90% CI	Item 90% CI	Seeds $\lvert\Delta\rvert \leq 5$	Verdict
Extraction	97.50	96.95	−0.55	[−2.07, +0.98]	[−1.80, +0.86]	8/8	Equivalent
Json	98.75	95.23	−3.52	[−4.56, −2.47]	[−5.31, −1.95]	7/8	Not equivalent
Router	93.75	63.59	−30.16	[−35.65, −24.66]	[−34.92, −25.47]	0/8	Not equivalent

The alignment tax is real, graded, and task-selective: undetectable (extraction, $\Delta = -0.55$), small but genuine (json, $\Delta = -3.52$), catastrophic (router, $\Delta = -30.16$). The gradient is directionally consistent with each task's format-alignment to the preference data across all 8 seeds—the task whose terse enum format is most orthogonal to the dominant verbose style axis (router) degrades most, the task with matching tool_calling coverage (json) degrades modestly, and the orthogonal-but-non-antagonistic format (extraction) is preserved. With only three tasks, we report this co-occurrence without claiming the causal link is established. Extraction also lacks direct preference coverage yet is preserved, so coverage absence alone does not predict degradation. Json has 200 matching tool_calling pairs yet still degrades; the per-axis dynamics (§4.3) explain why—tool_calling coverage provides partial but not complete protection against the dominant style-axis aggregate pressure.

4.2 Non-Monotonic Dynamics with Two Distinct Processes

Router's degradation is not monotonic in margin. Decomposing router accuracy into its two failure modes at each checkpoint (Table 2) reveals two processes: FORMAT_BREAK rate is non-monotonic, peaking at margin ≈3 (0.196) then partly subsiding by final (0.106); content-error rate rises monotonically (0.148 → 0.258), never recovering. The partial recovery in overall accuracy after margin3 is driven by format-break subsiding, not by content correctness improving.

Table 2: Router trajectory decomposition (mean ± std over 8 seeds; nf4, training backend).

Checkpoint	Margin	Semantic acc	FORMAT_BREAK rate	Content-error rate
step20	0.75	0.852 ± 0.015	0.001 ± 0.002	0.148 ± 0.016
margin1	1.09	0.816 ± 0.030	0.007 ± 0.010	0.177 ± 0.038
margin3	3.12	0.587 ± 0.061	0.196 ± 0.089	0.217 ± 0.071
margin5	5.13	0.626 ± 0.049	0.133 ± 0.096	0.241 ± 0.083
final	6.73	0.636 ± 0.077	0.106 ± 0.070	0.258 ± 0.074

Format breaks at margin ≈3 take the form of invented UPPER_SNAKE_CASE pseudo-labels derived from query content (e.g., TOTAL_REVENUES_2023_Q2, REPORT_PER_ANALYSIS). The valid router labels are also UPPER_SNAKE_CASE, so the model generates in the correct register but invents labels outside the valid 5-category set—a label-vocabulary interference specific to the preference data's financial domain, not degeneration, truncation, or style bleed. This replicates across matrix seeds.

The margin ≈3 instability is not limited to router. On extraction, the SURFACE_FORM register-shift rate mirrors the non-monotonic shape: it spikes at margin ≈3 (mean 0.37, range 0.03–0.77) then subsides to 0.09 by final. Under the strict scorer this register shift counts as a capability collapse (extraction accuracy 0.550 at margin3, seed0); under scorer v2's canonicalization the same checkpoint scores 0.981 (seed0)—96% of apparent errors are surface-form shifts ("Item 1" as "1"), not semantic failures. This is the paper's most concrete demonstration of why artifact-aware scoring is necessary.

Router accuracy reaches its trajectory minimum at margin3 on 6 of 8 seeds (mean 0.587, range 0.444–0.650; two seeds drift slightly lower afterward). Degradation begins before the margin 1–3 interval: step20 already shows −8.6 pts from baseline (28% of the net −30.2 pt loss), and margin1 accounts for −12.2 pts (40%). The checkpoint resolution cannot pinpoint the trough more precisely than "between margin ∼1 and ∼3." Despite consistent trough depth, recovery is highly seed-dependent: endpoint accuracy ranges 0.488 (seed5, near-zero recovery) to 0.769 (seed1), and the final format-break rate ranges 0.019–0.256. The damage is reproducible; the recovery is stochastic. Figure 1 shows per-seed margin-vs-capability trajectories: extraction forms a tight band near baseline, json shows mid-trajectory volatility but a narrow endpoint range, and router forms a wide fan—the variance structure itself communicates the finding.

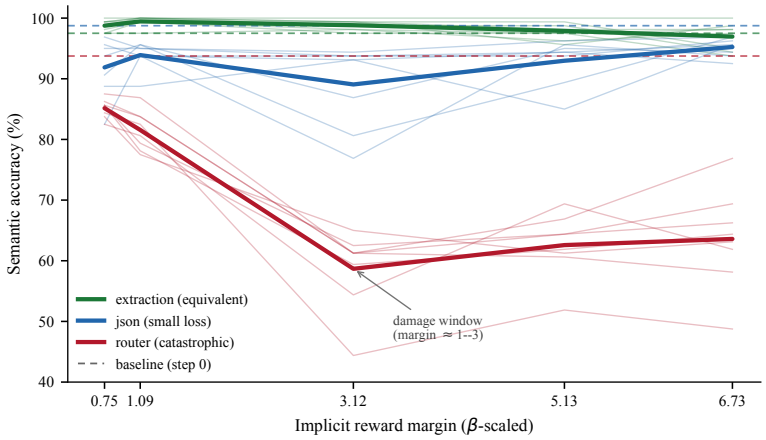


Figure 1: Per-seed margin-vs-capability trajectories for all three tasks (nf4, training backend). Thin lines are individual seeds; thick lines are the 8-seed mean; dashed lines are per-task baselines. Extraction (green) stays near baseline; json (blue) is volatile mid-trajectory but converges narrowly; router (red) fans out, with the trough at margin ≈3 on 6/8 seeds.

4.3 Mechanism: Rejected Collapse and Axis-Dependent Backend Sensitivity

Decomposing the implicit reward (policy log-prob minus reference log-prob) into chosen and rejected components shows chosen reward is approximately stationary at the final checkpoint under both backends (nf4: $+0.052$; fp16: -0.028), consistent with the SFT retention term (weight 0.5) anchoring chosen-completion likelihood. Under fp16, chosen dips to -0.173 at margin3 before recovering—a transient decline in Rafailov et al.'s predicted direction that the retention term dampens but does not fully block; nf4 shows a $2.4\times$ shallower dip at the same checkpoint (-0.072 vs -0.173). Rejected reward falls monotonically under both backends; chosen's contribution is $\leq 4\%$ in magnitude at every checkpoint, so the full margin growth is accounted for by rejected-completion collapse alone. The retention term protects chosen but provides no protection to rejected, so DPO's gradient pressure is channeled entirely into rejected suppression. Even with chosen explicitly anchored, the behavioral alignment tax remains—router -30 points, json one-directional loss—driven entirely by the rejected-suppression channel the retention term does not protect (implications for Cal-DPO and Rafailov et al. in §5.3). Figure 2 (nf4, training backend) shows the per-seed trajectories.

The per-axis decomposition reveals that per-pair ordering is not backend-invariant. Table 3 reports per-axis means at the final checkpoint under both backends.

Table 3: Per-axis reward decomposition at final checkpoint (8-seed mean; 549 training pairs: 378 style, 171 tool_calling).

Axis	Pairs	Margin (nf4)	Margin (fp16)	r_{ch} (nf4)	r_{rej} (nf4)	r_{ch} (fp16)	r_{rej} (fp16)
Style	378	8.119	2.466	$+0.012$	-16.226	-0.024	-4.955
Tool_calling	171	4.923	4.021	$+0.142$	-9.704	-0.038	-8.080

Under nf4 (the training backend—the margins the optimizer actually experienced), style dominates via both higher per-pair margin (8.12 vs 4.92) and pair count (aggregate: 3069 vs 842, $3.65\times$). Under fp16 (re-measurement), the per-pair ordering reverses: tool_calling is higher (4.02 vs 2.47), though style still produces more aggregate margin through pair count (932 vs 688, $1.36\times$). The divergence is concentrated in the style axis's rejected completions (nf4 -16.2 vs fp16 -5.0; tool_calling -9.7 vs -8.1).

Under both backends, style produces more aggregate margin. The behavioral gradient is consistent with this aggregate pressure pattern—the task most format-orthogonal to the verbose style axis (router) absorbs the most damage.

This per-pair distinction is tested on held-out data. On 98 held-out pairs (69 style, 29 tool_calling), tool_calling margin exceeds style margin at every checkpoint under fp16 (8/8 seeds; Appendix F,

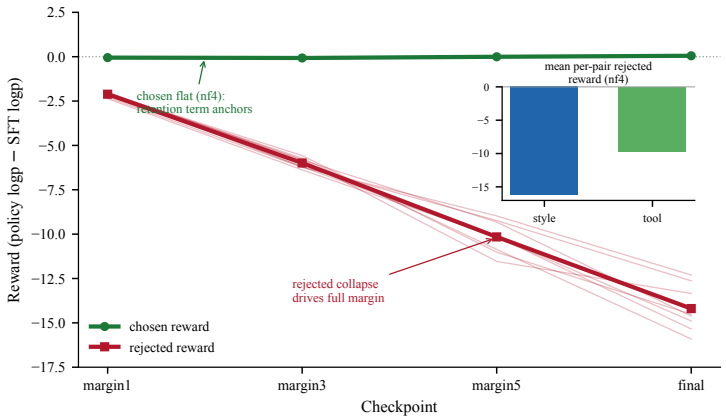


Figure 2: Reward decomposition into chosen and rejected components across checkpoints (nf4, training backend; thin lines per seed, thick lines 8-seed mean). Chosen reward stays flat near zero (retention term); rejected reward collapses, accounting for the full margin growth. Inset: mean per-pair rejected reward (nf4) at the final checkpoint (style vs tool_calling).

Table 13). Under nf4, tool_calling leads at margin1–3 (8/8 seeds; Table 14) but style overtakes by margin5, consistent with the axis-dependent inflation measured on training pairs at final. The pre-registered prediction that style margin would exceed tool_calling margin at every checkpoint (Note 2, secondary #3) is falsified under both backends: the prediction holds at 0/4 checkpoints under fp16 and 2/4 under nf4. The final-checkpoint per-pair ordering is backend-dependent.

4.4 Gold-Logp Dynamics Are Task-Selective

We next examine whether gold completions on evaluation items are stable and whether the pattern generalizes across tasks. All Δ_{gold} values are computed in fp16 on Mac; adapters were trained under nf4 on T4, but adapter weights are stored in full precision, so loading nf4-trained adapters onto an fp16 base is valid. Within-backend comparisons are internally consistent; absolute magnitudes may differ from nf4.

Table 4: Same-backend fp16 Δ_{gold} (gold_logp(final) − gold_logp(SFT)), 8-seed mean.

Task	Δ_{gold}	95% CI	Behavioral Δ
Extraction	-0.15	$[-0.18, -0.13]$	-0.5 pts (equivalent)
Json	-1.04	$[-1.20, -0.89]$	-3.5 pts (not equivalent)
Router	-1.47	$[-1.62, -1.35]$	-30 pts (not equivalent)

Gold deflation is task-selective (Table 4): extraction deflates modestly (-0.15 nats/token), json and router substantially (-1.04, -1.47). The ordering—extraction least, json intermediate, router most—is directionally consistent with the behavioral gradient across all 8 seeds, with non-overlapping CIs between tasks. The relationship is highly non-linear, however: router's gold deflates only $1.4\times$ more than json's, but its behavioral degradation is $8.6\times$ worse. Gold-logp magnitude alone does not explain the behavioral gradient. Json's substantial deflation (-1.04) paired with only minor behavioral loss (-3.5 pts) suggests that structured output formats with redundant constraints (typed fields, function signatures) may maintain correct greedy-decode behavior despite large probability-mass redistribution—a task-structural buffer router's terse single-token enum format does not provide.

The task-selective ordering is robust across quantization backends. Using the matrix's nf4 records (step20→final), the ordering extraction $\ll$ json $<$ router is identical (nf4 Δ_{gold}: extraction -0.008, json -0.22, router -1.59; non-overlapping CIs; Appendix G). Absolute magnitudes diverge—extraction's deflation nearly vanishes under nf4 ($\sim 18\times$), json compresses $\sim 5\times$, router is essentially unchanged—reflecting both the quantization difference and the different baselines; whether this reflects quantization-dependent sensitivity, baseline differences, or an interaction is unresolved. The task-selective ordering—the load-bearing claim—holds under both backends.

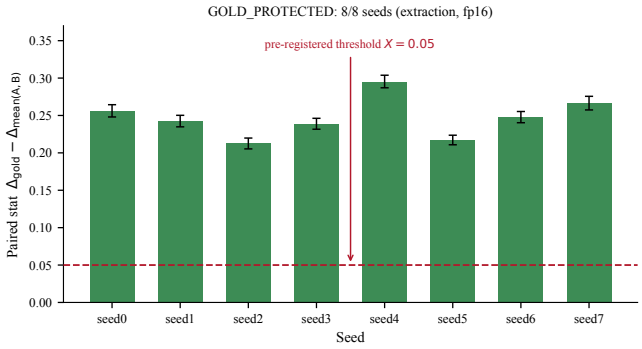


Figure 3: Paired GOLD_PROTECTED statistic per seed with 95% CI (extraction, fp16). All 8 bars clear the pre-registered threshold $X = 0.05$ (dashed).

On extraction we can go further. Teacher-forcing gold alongside control A (different section identifier) and control B (different ticker) through each seed's final checkpoint yields the paired GOLD_PROTECTED statistic (Table 5, Figure 3). All 8 seeds return GOLD_PROTECTED: every stat exceeds the pre-registered threshold ($X = 0.05$) with CIs far from zero. Gold is selectively protected—it declines modestly (-0.10 to -0.23 nats/token) while incorrect controls collapse $2.1\times$ to $3.5\times$ more (-0.34 to -0.47); the ordering of correct over incorrect completions is preserved and strengthened. Because the controls are extraction-specific, whether gold is relatively protected on json and router is untested (Table 4 reports their absolute Δ_{gold} only). Neither probe seed shows the catastrophic CHES failure mode: control-C rejected-register text falls sharply on both probe seeds where it was measured (-4.6 to -5.1 nats/token; control-C was not in the matrix or Mac runs). Finally, KL(policy $\|$ reference) on extraction gold completions rises from 0.022 at margin1 to 0.082 at final, confirming mild policy drift that plateaus as margin grows from $\sim$5 to $\sim$7 (Appendix G); KL is computed on extraction gold only, so drift on router may be larger.

Table 5: Paired GOLD_PROTECTED statistic (fp16, extraction, 160 items per seed).

Seed	Stat	95% CI	Δ_{gold}	Δ_{ctrl}
0	$+0.256$	$[+0.248, +0.264]$	-0.102	-0.358
1	$+0.242$	$[+0.235, +0.250]$	-0.226	-0.469
2	$+0.213$	$[+0.205, +0.220]$	-0.147	-0.360
3	$+0.239$	$[+0.232, +0.246]$	-0.102	-0.341
4	$+0.295$	$[+0.287, +0.304]$	-0.150	-0.446
5	$+0.217$	$[+0.211, +0.224]$	-0.194	-0.411
6	$+0.248$	$[+0.240, +0.255]$	-0.138	-0.386
7	$+0.267$	$[+0.257, +0.276]$	-0.168	-0.435

4.5 Backend Sensitivity of QLoRA Margins

Implicit reward margins under QLoRA differ up to $3.4\times$ between nf4 and fp16 re-measurement on the same adapters (the same SFT adapter as reference, scored within each backend). The divergence is axis-dependent: style-axis margins inflate $3.3\times$ (train) to $3.4\times$ (held-out), while tool_calling inflates $1.2\times$ on both splits. Style-rejected completions are long degraded-text sequences, tool_calling-rejected are short tool-swaps; length and content co-vary with axis, and the cause of the differential inflation is not isolated. Final-checkpoint chosen anchoring is backend-robust; the mid-training trajectory is not—the margin3 transient is $2.4\times$ deeper under fp16 (-0.173 vs nf4's -0.072).

This is directionally opposite in our measurements to the gold-logp nf4/fp16 discrepancy (§4.4), where nf4 magnitudes are smaller for extraction ($18\times$) and json ($5\times$) and comparable for router—though the gold-logp comparison is baseline-confounded (SFT vs step20) while the margin comparison uses the same SFT adapter scored within each backend, making the contrast suggestive, not established. Chosen-flatness at the final checkpoint and aggregate ordering (style $>$ tool under both backends) are backend-invariant; per-pair ordering is not. Margin-based monitoring of

QLoRA-trained models is therefore backend-sensitive; per-pair and per-axis comparisons require same-backend measurement.

5 Discussion

5.1 A Two-Level Mechanistic Account

The behavioral and probability-layer findings form a coherent—though not causally proven—account. At the training level, the retention term anchors chosen-completion likelihood and channels DPO's pressure entirely into rejected suppression (§4.3); the style axis dominates in aggregate under both backends—consistent with why the task most format-orthogonal to the verbose style axis (router) absorbs the most damage while the task with direct tool_calling coverage (json) absorbs less—whereas per-pair ordering is backend-dependent. At the probability level, gold-logp deflation is task-selective and directionally consistent with the behavioral gradient; on extraction—the best-preserved task, whose gold completions overlap most with training-pair chosen completions (§3.3)—gold is selectively protected relative to incorrect controls (8/8 seeds), consistent with Ren and Sutherland's squeezing effect concentrating mass onto gold tokens that share entity vocabulary with training pairs, an effect router's dissimilar terse format may not enjoy. Both backends' chosen minima sit at margin3 (behavioral metrics are nf4-generated; the fp16 chosen dip is a re-measurement, not the source of the behavioral instability), coinciding with the behavioral instability window (FORMAT_BREAK 0.196, SURFACE_FORM 0.37); no causal claim is made.

The two levels do not form a single causal chain: gold-logp magnitude is non-linearly related to behavior (router's gold deflates $1.4\times$ more than json's but degrades $8.6\times$ worse), and the behavioral gradient is better explained by format-alignment to the style-axis aggregate pressure than by gold-logp dynamics. We present the two levels as complementary—rejected collapse explains which tasks are pressured; gold-logp dynamics describe how probability mass redistributes—without claiming one causes the other.

5.2 Falsification Chronology

The pre-registered plan forced three falsifications and two refinements, each producing a more precise characterization than the hypothesis it replaced; three further hypotheses were abandoned during design (a confounded causal-clamp experiment, router-as-primary after a non-monotonic transient appeared, and a targeted-displacement probe that pointed to global drift; Appendix C).

Three hypotheses were falsified outright. First, the bounded-null headline (all three tasks preserved within ± 5) was rejected when the falsifier fired on json and router (§4.1). Second, an apparent extraction collapse at margin ≈ 3 (accuracy 0.550, seed0) was traced to a scorer artifact—canonicalized scoring gives 0.981 (seed0), 96% of apparent errors being surface-form register shifts; the measurement was killed, not refined. Third, the pre-registered prediction that style-axis held-out margin would exceed tool_calling margin at every checkpoint is falsified under both backends (fp16: 0/4 checkpoints; nf4: 2/4), though the final-checkpoint per-pair ordering is backend-dependent (style $>$ tool under nf4, tool $>$ style under fp16). The falsification reveals that per-pair ordering is not backend-invariant; aggregate style dominance holds under both.

Two characterizations were refined. The pilot's "margin-independent early drop" on router became a damage window at margin ≈ 1–3 with two distinct processes (transient format-break + monotonic content-error) followed by stabilization (§4.2). And the initial "gold barely moves" characterization (nf4, $|\Delta_{\text{gold}}| \leq 0.01$) was refined to "gold is selectively protected but task-selectively deflates" once fp16 same-backend measurement revealed substantial json and router deflation (§4.4). Each pre-registered gate forced a more precise account; the result is more nuanced than any single prior framing predicted.

5.3 Engagement with Prior Work

Our data engages three displacement-literature claims, with the caveat that our objective includes an SFT retention term (the RPO formulation; Liu et al. [6]; weight 0.5) on chosen completions—making it DPO + retention, not vanilla DPO. *Rafailov et al.* predict from their Q-function framework that chosen log-prob should decrease under DPO from an SFT-initialized model; our retention

term directly counteracts this. The chosen-flatness we observe (-0.072 to $+0.052$ over 7 units of margin growth) is therefore at least partly a consequence of the training objective, not purely an empirical finding about DPO dynamics; it confirms the retention term is effective in this regime but cannot speak to whether chosen would decline under standard DPO without retention. The fp16 re-measurement reveals a transient mid-training chosen dip (-0.173 at margin3, recovering to -0.028 by final)—a residual decline in the direction Rafailov et al. predict, dampened but not fully blocked by the retention term; under nf4 the dip is $2.4\times$ shallower (-0.072). The retention term largely counteracts the predicted decline; dequantized measurement reveals the residual. *Cal-DPO* adds a calibration term to prevent chosen-reward decline; our retention term achieves a similar anchoring by a different mechanism. That chosen is anchored at the endpoint does not demonstrate Cal-DPO's concern is unfounded—it shows the concern is largely addressed; even with chosen anchored, the behavioral tax remains, driven entirely by the unprotected rejected-suppression channel. *IPO* was designed to bound the margin, treating unbounded growth as the core pathology; our margin grows to $\sim$7 with graded, format-selective damage rather than uniform catastrophe. This does not invalidate IPO; it characterizes a regime where damage is concentrated in the margin 1–3 window and is format-selective.

5.4 Format-Alignment: What It Explains and What It Does Not

The behavioral gradient (extraction preserved $\rightarrow$ json small loss $\rightarrow$ router catastrophic) is directionally consistent with format-alignment to the preference data's style axis across all 8 seeds, but with only three tasks we report this co-occurrence without claiming a causal mechanism. Two limits deserve attention. First, extraction has zero direct preference coverage yet is preserved, so coverage absence alone does not predict degradation; the distinguishing factor may be format-orthogonality, domain overlap (§3.3), or the retention term anchoring chosen completions that share extraction's entity vocabulary—all three co-vary and cannot be disentangled with three tasks. Second, json has 200 on-target tool_calling pairs yet degrades slightly: tool_calling coverage protects against the correctness axis but not against the dominant style-axis aggregate pressure, so the damage mechanism and the protection mechanism operate on different axes. Json's substantial gold deflation (-1.04) with only minor behavioral loss (-3.5 points) suggests redundant-constraint formats may buffer greedy-decode behavior against probability-mass redistribution; but extraction is also structured JSON yet deflates far less (-0.15), so format redundancy alone does not explain the difference—format-alignment co-varies. With two structured tasks and two co-varying variables, this hypothesis is illustrative, not testable.

5.5 Practical Implications

For practitioners running length-normalized DPO on small models with structured-output tasks: (1) monitor format-break rate at margin $\approx$3, not just endpoint accuracy—the transient spike is where damage concentrates and may be invisible at the final checkpoint; (2) audit preference-data composition for style-axis aggregate dominance—the behavioral gradient is consistent with aggregate training pressure under both backends, though per-pair ordering is backend-dependent and per-axis comparisons require same-backend measurement; (3) consider including an SFT retention term—anchoring is endpoint-level; mid-training chosen can dip transiently under faithful-precision measurement, though the no-retention counterfactual is untested.

5.6 Limitations

This study operates within explicit scope conditions: sparse LoRA on 8/32 layers (0.034% of parameters; not full fine-tuning), a single 4B model, length-normalized rewards with an SFT retention term (weight 0.5), easily-separable synthetic preferences, and three structured exact-format tasks. We do not study safety or refusal tasks, where Razin et al. demonstrated catastrophic displacement. Under naturalistic human preferences with higher chosen-rejected similarity (high CHES scores), the displacement dynamics Razin et al. characterize may produce different outcomes. Results may not generalize beyond these conditions.

Three pre-registered secondaries (#3, #4, #8) were computed post-hoc from saved adapters rather than logged during training: #4 under nf4 (Kaggle T4) with an fp16 re-measurement (Mac); #3 under fp16 (Mac) and nf4 (Kaggle T4); #8 under fp16 (Mac). Separately, the paired GOLD_PROTECTED statistic is extraction-only by design (controls A/B are task-specific). The format-alignment mechanism rests on $n = 3$ tasks—insufficient to distinguish format-orthogonality from coverage, domain

overlap, or task difficulty. The checkpoint schedule cannot pinpoint the damage window more precisely than "between margin $\sim$1 and $\sim$3." Gold-logp absolute magnitudes differ substantially between fp16 and nf4; the task-selective ordering is robust across both, but absolute values are not directly comparable (§4.4, §4.5, Appendix G).

6 Conclusion

The alignment tax under length-normalized DPO with an SFT retention term is real but graded and task-selective, with router's catastrophic loss temporally structured—with onset at step20 (-8.6 pts, 28% of net loss) and concentrated in a margin 1–3 damage window where transient format-break and persistent content-error operate as distinct processes. The retention term anchors chosen completions while DPO's gradient pressure channels entirely into rejected-completion suppression, producing a behavioral gradient that tracks format-alignment to the preference data's style axis more than margin magnitude. Artifact-aware scoring corrected a false-collapse signal, and pre-registered TOST equivalence testing prevented both false-positive and false-negative capability claims; a disclosed chronology of three falsified hypotheses and two refined characterizations documents how pre-registration forced successive refinement of initially simpler claims. That margin growth to $\sim$7 β-scaled units (nf4 training backend) produces graded rather than uniform degradation offers a differentiated picture—preservation where format-alignment is maintained, substantial risk where it is not—and a methodological template for future alignment-tax studies. These findings hold under specific scope conditions—sparse LoRA on 8/32 layers, a single 4B model, length-normalized rewards with an SFT retention term, and easily-separable synthetic preferences—and may not transfer to full fine-tuning, naturalistic preferences, or safety-critical tasks.

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Appendices

A Full Per-Seed TOST Data

Complete data behind Table 1. Per-task semantic accuracy (%) at each checkpoint, all 8 seeds, plus baseline_v2. $\Delta = \text{final} - \text{baseline}$ (percentage points). The TOST per-seed equivalence band is $|\Delta| \leq 5$.

Table 6: extraction — per-seed semantic accuracy (%).

Seed	Baseline	step20	margin1	margin3	margin5	final	Δ	TOST
0	97.50	97.50	97.50	98.12	96.25	97.50	+0.00	PASS
1	97.50	98.75	100.00	98.75	98.75	94.38	−3.12	PASS
2	97.50	99.38	99.38	99.38	98.12	98.12	+0.62	PASS
3	97.50	98.12	99.38	98.12	97.50	98.12	+0.62	PASS
4	97.50	99.38	100.00	99.38	95.62	98.75	+1.25	PASS
5	97.50	100.00	100.00	99.38	99.38	95.00	−2.50	PASS
6	97.50	97.50	99.38	97.50	97.50	93.75	−3.75	PASS
7	97.50	99.38	100.00	100.00	100.00	100.00	+2.50	PASS

Table 7: json — per-seed semantic accuracy (%).

Seed	Baseline	step20	margin1	margin3	margin5	final	Δ	TOST
0	98.75	82.50	93.75	93.12	94.38	92.50	−6.25	FAIL
1	98.75	93.75	95.62	80.62	89.38	96.88	−1.88	PASS
2	98.75	88.75	88.75	93.12	85.00	95.62	−3.12	PASS
3	98.75	96.88	95.00	93.75	93.75	96.25	−2.50	PASS
4	98.75	91.88	93.75	76.88	95.62	93.75	−5.00	PASS
5	98.75	95.00	95.00	94.38	96.25	96.88	−1.88	PASS
6	98.75	90.62	95.62	86.88	95.00	94.38	−4.38	PASS
7	98.75	95.62	93.75	93.75	94.38	95.62	−3.12	PASS

Table 8: router — per-seed semantic accuracy (%).

Seed	Baseline	step20	margin1	margin3	margin5	final	Δ	TOST
0	93.75	85.62	78.12	59.38	61.88	64.38	−29.37	FAIL
1	93.75	87.50	86.88	61.25	66.88	76.88	−16.87	FAIL
2	93.75	86.25	83.75	61.25	60.62	58.13	−35.62	FAIL
3	93.75	83.75	77.50	65.00	61.25	63.12	−30.63	FAIL
4	93.75	82.50	80.62	54.37	69.38	61.88	−31.87	FAIL
5	93.75	84.38	82.50	44.38	51.88	48.75	−45.00	FAIL
6	93.75	85.62	83.75	61.25	64.38	69.38	−24.38	FAIL
7	93.75	85.62	79.38	62.50	64.38	66.25	−27.50	FAIL

Per-seed TOST pass/fail is shown in the final column; item-level bootstrap 90% CIs at the final checkpoint are reported per task in Table 1. The router failure-mode decomposition across the checkpoint trajectory is in main-text Table 2; Figure 1 visualizes the same trajectory per seed.

B Scorer v2 Rules and Artifact Walkthrough

Scorer architecture. Scorer v2 applies a deterministic classification to each generated output: (1) **FORMAT_BREAK** — output cannot be parsed by the task-specific format gate (router: a bare string matching one of the five valid labels; json/extraction: valid JSON with required keys); (2) **PASS** — parsed output matches the gold answer after canonicalization; (3) **SURFACE_FORM** — parsed output contains the correct semantic content but differs in surface form (e.g., "Item 1" vs "1"); (4) **SEMANTIC_ERROR** — parsed, well-formed output with incorrect content. Semantic accuracy counts PASS + SURFACE_FORM as correct; FORMAT_BREAK and SEMANTIC_ERROR are failures.

Canonicalization rules (extraction). The section-identifier field is canonicalized before comparison: extract the numeric portion via regex `\d+[A-Z]?` ("Item 7A" $\rightarrow$ "7A", "Item 1" $\rightarrow$ "1"), case-fold the alpha suffix, and ignore appended titles ("Item 7 — Management's Discussion" $\rightarrow$ "7"). Ticker and filing_date use exact string match (case-insensitive for ticker).

The artifact: $0.550 \rightarrow 0.981$. At the margin3 checkpoint, extraction accuracy under strict string matching is 0.550 (seed0); under scorer v2's canonicalization the same checkpoint scores 0.981. The 69 items ($69/72 = 96\%$) that flip from FAIL to PASS are exclusively section-identifier surface-form variants ("1" where gold says "Item 1", "7A" where gold says "Item 7A"); the semantic content is correct in every case, with no ticker or filing_date errors. The 8-seed mean is $0.616 \rightarrow 0.988$. The register-shift rate (SURFACE_FORM tier) peaks at margin $\approx$3 (mean 0.37 across 8 seeds) then subsides to 0.09 by final, mirroring router's FORMAT_BREAK non-monotonic trajectory. The canonicalization rules were frozen before the confirmatory matrix; no rules were added or modified after observing matrix results.

C Pre-Registration Reconciliation and Falsification Chronology

C.1 Secondary reconciliation. Every pre-registered secondary from Note 2 (frozen 2026-06-18) is accounted for below.

Table 9: Pre-registered secondary reconciliation.

#	Secondary	Status	Location
1	Trajectory-min per seed/task	Reported	§4.2 (router trough at margin3, 6/8)
2	FORMAT_BREAK trajectory	Reported	§4.2 (Table 2)
3	Held-out margin per category	Reported — **prediction falsified**	§4.3 (Table 13)
4	Chosen/rejected reward decomposition	Reported (post-hoc GPU inference)	§4.3, Fig. 2 (per-axis Table 3)
5	Gold-logp trajectory $\times$ 3 tasks	Reported (fp16, 8 seeds)	§4.4 (Table 4)
6	Probe statistic replication (paired)	Extraction 8/8 GOLD_PROTECTED; json/router not computable (extraction-specific controls)	§4.4 (Table 5)
7	Register-shift tracking	Reported (SURFACE_FORM rate)	§4.2
8	KL-from-reference per checkpoint	Reported (post-hoc, fp16, 8 seeds)	§4.4 (Appendix G)

Secondaries #4 and #8 were not logged during the matrix training run: #4 was computed via a dedicated GPU inference run on saved adapters (Kaggle T4); #8 via Mac inference on saved adapters (fp16, proper token-level KL). #3 was computed via Mac inference on saved adapters. All three use the same checkpoint adapters as the matrix; no retraining was involved. The primary TOST analysis (§4.1) was unaffected.

C.2 Pre-registration deviations. *Checkpoint mechanism:* the pre-registered design (Note 2) specified fixed-step checkpoints with margin as a post-hoc label; the implementation used margin-crossing triggers for intermediate checkpoints (margin1: steps 24–30, margin3: 84–98, margin5:

112–138). Step20 and final (step 180) are fixed and identical across seeds; the primary endpoint is unaffected; cross-seed averages at intermediate checkpoints pool seeds at matched margin but different optimizer steps. *Early-stop configuration:* the design specified `max_opt_steps=180` and `no_early_stop=True`, enforced at runtime in `matrix_kernel.py` and `run_matrix.py`; all 8 seeds terminated at exactly step 180. However, `train_config.json` records module-level constants only and does not serialize these runtime parameters, which may create a misleading impression when inspecting saved artifacts. The execution matched the pre-registration; the config-file serialization did not. *Backend:* the pre-registration did not specify a quantization backend for any secondary. Secondary #3 (held-out per-category margin) was computed under fp16 (Mac) and nf4 (Kaggle T4); the per-pair ordering is backend-dependent at the final checkpoint (§4.3).

C.3 Falsification chronology. *Design phase (before confirmatory matrix):* (a) a causal-clamp experiment was designed but dropped as confounded—clamping the margin also alters the KL penalty and total optimization pressure, making "margin magnitude → capability loss" untestable; (b) router was initially the primary endpoint but moved to a secondary with trajectory decomposition after pilot analysis revealed a non-monotonic transient; (c) a targeted-displacement hypothesis was reformulated as "selective deflation" after a preliminary probe found controls fell more than gold (global drift, not targeted displacement); (d) three exploration-phase numbers lacking on-disk artifacts were re-measured or dropped (80-item baselines re-derived from the 160-item suite; router baseline corrected 0.950 → 0.938; gate floors re-derived: extraction 0.006, router 0.200, json 0.0125); (e) the suite was expanded 80 → 160 items per task to improve TOST power and frozen before the matrix; (f) the GOLD_PROTECTED threshold ($X = 0.05$) was frozen before the $n = 12$ preliminary probe existed.

Matrix phase (confirmatory): **Falsified** — the bounded-null headline (json and router failed ± 5 TOST); the extraction collapse at margin ≈ 3 (0.550 → 0.981 under canonicalization; measurement killed, not refined); style margin > tool_calling on held-out (falsified under both backends: fp16 $0/4$, nf4 $2/4$; final-checkpoint per-pair ordering backend-dependent). **Refined** — "margin-independent early drop" → damage window at margin ≈ 1–3 with two distinct processes; "gold absolutely stable" ($|\Delta_{\text{gold}}| \leq 0.01$, nf4) → "gold selectively protected but task-selectively deflates" (fp16 revealed json -1.04, router -1.47, extraction -0.15).

D Suite Hygiene

Source files: `data_eval/track1_{router,json,extraction}.jsonl`; leakage reference `kaggle_upload/training_data/dpo_pytorch.jsonl`; degeneracy floors `kaggle_upload/audit_eval.py:91`; gold self-consistency via `kaggle_upload/scorer.py`.

Table 10: Suite hygiene properties.

Property	extraction	json	router
n items	160	160	160
Unique prompts	160	160	160
Duplicate prompts	0	0	0
Verbatim leakage (eval ∩ train)	0	0	0
Gold self-consistency (gold-as-output PASS)	160/160	160/160	160/160
Degeneracy floor (always-one-answer)	∼0.006	0.0125	0.20

Class balance. *router* (5 classes): COMPANY_DEEP_DIVE, FILING_ANALYSIS, FILING_DIFF, OUT_OF_SCOPE, SIMPLE_QUERY = 32 each. *json* (8 tool classes): analyst_ratings, company_metrics, company_news, diff_filing_sections, get_filing_section, insider_trades, search_filings, stock_price = 20 each. *extraction* (6 section classes): Item 1 = 27, Item 1A = 27, Item 3 = 26, Item 7 = 27, Item 7A = 26, Item 8 = 27.

Zero verbatim leakage: 0/160 eval prompts appear as a training `question` (exact-string match) for all three tasks. Gold self-consistency: 160/160 gold answers pass their own scorer. Zero duplicate prompts: 160 unique prompts / 160 items. Degeneracy floors (a constant always-one-answer policy): extraction 0.006, router 0.20 (5 balanced classes), json 0.0125 ($2/160$). Extraction golds are unique (constant-output floor $1/160 \approx 0.006$); router has 5 balanced classes ($32/160 = 0.20$); json has 18 duplicate tool-call tuples ($2/160 = 0.0125$, maximum multiplicity 2).

E Training Configuration and Hash Verification

SFT checkpoint: hash `29a3794b...585f02`, verified identical across all 8 seed runs; adapter weights loaded from `adapters/sft_pytorch/adapter_model.safetensors`. **Tokenizer and chat template:** byte-identical across all sources (SFT adapter, base model, eval harness), verified by SHA-256 hash comparison.

LoRA configuration: $r = 8$, $\alpha = 16$, dropout $= 0.05$; `target_modules` = [q_proj, k_proj, v_proj, o_proj] (explicitly configured, not peft default); bias = none; peft version 0.19.1. Injected layers: 3, 7, 11, 15, 19, 23, 27, 31 (the 8 full-attention layers in Qwen3.5-4B's hybrid architecture; 1.57M trainable parameters, 0.034% of 4.66B total).

Training hyperparameters: AdamW, lr $= 5 \times 10^{-5}$, constant schedule, weight_decay $= 0.01$; gradient clipping 1.0; batch size 8 (gradient accumulation); β (DPO) $= 0.5$; SFT retention weight $= 0.5$; max optimizer steps 180 (enforced via `matrix_kernel.py` runtime parameter); early stop disabled (`no_early_stop=True` via `run_matrix.py`); divergence guard: abort if NaN or smoothed margin > 20.

Config-file note: `train_config.json` records module-level constants and does not serialize the runtime parameters `max_opt_steps` and `no_early_stop`; runtime enforcement was applied in `matrix_kernel.py` and `run_matrix.py`, confirmed by source-code inspection. All 8 seeds terminated at exactly step 180. **Checkpoint adapters on disk:** 8 seeds $\times$ 5 checkpoints $= 40$ adapter directories, all verified present with complete `adapter_model.safetensors` files.

F Reward Decomposition and Held-Out Margin Per-Seed

Source: `reward_decomposition/summary.json` (β=0.5, reference = sft_baseline). r = policy_logp $-$ sft_logp (length-normalized); margin $= \beta \cdot (r_{\text{chosen}} - r_{\text{rejected}})$. The 8-seed-mean per-axis decomposition under both backends is in main-text Table 3; per-seed nf4 values follow.

Table 11: F.1 — r_{chosen}, r_{rejected}, margin per checkpoint, per seed (nf4).

Seed	Ckpt	r_{chosen}	r_{rejected}	margin
0	margin1	−0.044	−2.087	1.021
0	margin3	−0.094	−5.569	2.738
0	margin5	−0.001	−11.016	5.508
0	final	0.051	−14.518	7.285
1	margin1	−0.082	−2.365	1.141
1	margin3	−0.060	−6.380	3.160
1	margin5	0.006	−10.146	5.076
1	final	0.055	−15.344	7.699
2	margin1	−0.039	−2.075	1.018
2	margin3	−0.063	−5.714	2.825
2	margin5	−0.033	−10.861	5.414
2	final	0.052	−15.909	7.981
3	margin1	−0.045	−2.024	0.990
3	margin3	−0.052	−5.822	2.885
3	margin5	−0.033	−9.320	4.644
3	final	0.060	−14.615	7.337
4	margin1	−0.054	−2.285	1.116
4	margin3	−0.074	−6.235	3.080
4	margin5	0.013	−8.973	4.493
4	final	0.054	−12.302	6.178
5	margin1	−0.047	−2.014	0.984
5	margin3	−0.100	−6.196	3.048
5	margin5	0.018	−9.233	4.626
5	final	0.059	−12.631	6.345
6	margin1	−0.051	−2.005	0.977
6	margin3	−0.075	−5.799	2.862
6	margin5	−0.023	−11.535	5.756
6	final	0.057	−13.346	6.701
7	margin1	−0.051	−2.052	1.001
7	margin3	−0.057	−6.206	3.074
7	margin5	0.020	−10.166	5.093
7	final	0.032	−14.890	7.461

Table 12: F.2 — Per-axis margin at final checkpoint (style vs tool_calling), per seed (nf4).

Seed	style r_{ch}	style r_{rej}	style margin	tool r_{ch}	tool r_{rej}	tool margin
0	0.023	−16.560	8.291	0.113	−10.005	5.059
1	0.013	−17.726	8.870	0.147	−10.077	5.112
2	0.013	−18.820	9.417	0.139	−9.475	4.807
3	0.019	−16.696	8.358	0.149	−10.016	5.082
4	0.013	−13.536	6.774	0.147	−9.572	4.860
5	0.018	−14.142	7.080	0.149	−9.292	4.720
6	0.016	−15.118	7.567	0.148	−9.428	4.788
7	−0.019	−17.206	8.594	0.144	−9.770	4.957

Table 13: F.3 — Held-out margin by category per checkpoint, per seed (fp16; 98 pairs: 69 style, 29 tool_calling; β=0.5). Under fp16, tool_calling margin exceeds style margin at every checkpoint on every seed.

Seed	Ckpt	style margin	tool_calling margin
0	margin1	0.650	1.301
0	margin3	1.738	2.664
0	margin5	2.480	3.253
0	final	2.501	4.034
1	margin1	0.698	1.484
1	margin3	1.843	3.069
1	margin5	2.241	3.383
1	final	2.476	3.905
2	margin1	0.615	1.361
2	margin3	1.729	2.961
2	margin5	2.498	3.285
2	final	1.904	3.709
3	margin1	0.671	1.189
3	margin3	1.987	2.743
3	margin5	2.989	3.184
3	final	2.230	4.002
4	margin1	0.721	1.411
4	margin3	1.882	2.895
4	margin5	2.044	3.270
4	final	2.115	3.856
5	margin1	0.595	1.230
5	margin3	1.977	2.782
5	margin5	2.225	3.230
5	final	1.998	3.814
6	margin1	0.680	0.952
6	margin3	1.859	2.777
6	margin5	2.568	3.297
6	final	1.998	3.739
7	margin1	0.706	1.077
7	margin3	2.055	2.913
7	margin5	2.490	3.505
7	final	3.354	3.908

Table 14: F.4 — Held-out per-category margin under nf4 (8-seed mean; 98 pairs: 69 style, 29 tool_calling). Tool_calling leads through margin3; style overtakes by margin5 under axis-dependent nf4 inflation.

Checkpoint	Style (nf4)	Tool (nf4)
margin1	0.780	1.466
margin3	2.620	3.426
margin5	5.435	4.020
final	7.919	4.753

Table 15: F.5 — fp16 train reward decomposition, pooled per checkpoint (8-seed mean; 549 train pairs: 378 style, 171 tool_calling; β=0.5, reference = sft_baseline). Chosen dips transiently at margin3 (-0.173) before recovering; rejected falls monotonically.

Checkpoint	r_{chosen}	r_{rejected}	margin
margin1	-0.045	-1.785	0.870
margin3	-0.173	-4.661	2.244
margin5	-0.126	-5.743	2.809
final	-0.028	-5.928	2.950

G Gold-Logp Per-Seed Data, nf4 Cross-Check, and KL Trajectory

G.1 fp16 same-backend Δ_{gold} (per-seed). Source: `gold_3task_summary.json` (fp16_mac, reference = sft_baseline). Δ_{gold} = gold(final) − gold(sft), per-token length-normalized. Bracketed values are per-seed bootstrap 95% CI.

Table 16: G.1 — per-seed fp16 Δ_{gold} with 95% CI.

Seed	Extraction	Json	Router
0	-0.102 $[-0.108, -0.095]$	-1.505 $[-1.581, -1.430]$	-1.259 $[-1.645, -0.902]$
1	-0.226 $[-0.232, -0.220]$	-1.177 $[-1.281, -1.074]$	-1.616 $[-2.029, -1.226]$
2	-0.147 $[-0.153, -0.141]$	-1.001 $[-1.078, -0.925]$	-1.458 $[-1.903, -1.044]$
3	-0.102 $[-0.108, -0.096]$	-0.957 $[-1.057, -0.858]$	-1.305 $[-1.723, -0.919]$
4	-0.150 $[-0.157, -0.143]$	-0.906 $[-1.001, -0.812]$	-1.915 $[-2.406, -1.451]$
5	-0.194 $[-0.200, -0.188]$	-0.761 $[-0.854, -0.671]$	-1.307 $[-1.692, -0.939]$
6	-0.138 $[-0.145, -0.131]$	-1.175 $[-1.252, -1.101]$	-1.377 $[-1.749, -1.028]$
7	-0.168 $[-0.177, -0.160]$	-0.803 $[-0.904, -0.703]$	-1.539 $[-1.999, -1.109]$

Across-seed means (8 seeds): extraction -0.153 $[-0.182, -0.126]$, json -1.036 $[-1.204, -0.891]$, router -1.472 $[-1.624, -1.345]$.

G.2 nf4 cross-check. nf4 Δ_{gold} is computed as gold_logp(final) − gold_logp(step20), using the matrix's own records; no nf4 SFT baseline gold-logp was recorded, so step20 is the earliest available reference.

Table 17: G.2 — nf4 Δ_{gold} (step20→final), 8-seed mean.

Task	Δ_{gold} (nf4)	95% CI	Δ_{gold} (fp16) for comparison
Extraction	-0.008	$[-0.014, -0.003]$	-0.15
Json	-0.22	$[-0.34, -0.11]$	-1.04
Router	-1.59	$[-1.99, -1.19]$	-1.47

Table 18: G.2 — per-seed nf4 Δ_{gold} (step20→final).

Seed	Extraction	Json	Router
0	-0.0089	-0.1495	-1.6280
1	-0.0134	-0.5054	-1.6686
2	-0.0089	-0.1291	-1.0105
3	-0.0004	-0.3407	-1.7108
4	$+0.0002$	-0.1034	-1.9370
5	-0.0164	-0.1119	-2.4182
6	-0.0047	-0.2077	-0.9808
7	-0.0141	-0.2485	-1.3311

The task-selective ordering (extraction $\ll$ json $<$ router) is identical under both backends, with non-overlapping CIs. Absolute magnitudes diverge, reflecting both the quantization difference and the different baselines (fp16 uses the pure SFT adapter; nf4 uses step20, which has absorbed 20 DPO steps). The divergence is itself task-selective: extraction's deflation nearly vanishes under nf4 ($\sim$18$\times$ compression), json compresses $\sim$5$\times$, while router's magnitude is essentially unchanged.

G.3 GOLD_PROTECTED paired statistic (fp16, extraction, per-seed). Source: `gold_controls_summary.json` (fp16 unquantized, paired stat $\Delta_{\text{gold}} - \Delta_{\text{mean}(A,B)}$, reference = sft_baseline, X=0.05). GOLD_PROTECTED verdict when stat $\geq X$.

Table 19: G.3 — per-seed GOLD_PROTECTED statistic.

Seed	stat	95% CI	Δ_{gold}	Δ_{ctrl} (mean A,B)	Verdict
0	0.2561	$[0.2480, 0.2644]$	-0.1016	-0.3576	GOLD_PROTECTED
1	0.2425	$[0.2348, 0.2502]$	-0.2262	-0.4687	GOLD_PROTECTED
2	0.2125	$[0.2053, 0.2198]$	-0.1470	-0.3595	GOLD_PROTECTED
3	0.2389	$[0.2316, 0.2462]$	-0.1022	-0.3411	GOLD_PROTECTED
4	0.2953	$[0.2870, 0.3038]$	-0.1502	-0.4455	GOLD_PROTECTED
5	0.2172	$[0.2108, 0.2236]$	-0.1938	-0.4110	GOLD_PROTECTED
6	0.2477	$[0.2403, 0.2553]$	-0.1379	-0.3856	GOLD_PROTECTED
7	0.2666	$[0.2574, 0.2756]$	-0.1684	-0.4350	GOLD_PROTECTED

Aggregate: 8/8 seeds GOLD_PROTECTED.

G.4 KL(policy $\|$ reference) trajectory.

Table 20: G.4 — KL trajectory (extraction gold, fp16, 8-seed mean $\pm$ std).

Checkpoint	Mean KL	Std
margin1	0.022	0.005
margin3	0.057	0.017
margin5	0.083	0.014
final	0.082	0.008

KL rises monotonically through margin5 then plateaus ($0.083 \rightarrow 0.082$), confirming mild policy drift (all values < 0.1 nats/token) that saturates as margin continues to grow from $\sim$5 to $\sim$7. KL is computed on extraction gold completions only; drift on other tasks may be larger. Per-seed KL at final: seed0 $= 0.071$, seed1 $= 0.089$, seed2 $= 0.081$, seed3 $= 0.075$, seed4 $= 0.077$, seed5 $= 0.092$, seed6 $= 0.077$, seed7 $= 0.093$.

H Validation Margin Trajectories

Source: `track1_dpo_seed*_margin.jsonl`. `val_margin` is logged only at the evaluation cadence (`eval_every` $= 50 \rightarrow$ steps 50, 100, 150); it is not recorded at the checkpoint steps. The table reports every logged val_margin step.

Table 21: Validation margin at logged steps, with intermediate checkpoint steps for reference.

Seed	val@50	val@100	val@150	(margin1 step)	(margin3 step)	(margin5 step)
0	1.701	3.534	5.752	27	84	134
1	1.648	3.091	5.904	29	93	131
2	1.685	3.618	5.746	26	87	124
3	1.660	3.506	7.397	28	90	112
4	1.701	3.483	6.449	30	91	130
5	1.804	3.502	4.968	24	87	135
6	1.743	3.522	6.709	27	88	138
7	1.528	3.288	5.768	26	98	135

The 8-seed mean val_margin at step 150 is 6.087 (rounds to 6.1).

Val-subset note. The training-time `val_margin` logged in the margin files is computed over a fixed 20-pair subset (15 style, 5 tool_calling) of the 98 held-out pairs under the nf4 training backend—not the full 98-pair set. Post-hoc fp16 recomputation on the same 20 pairs yields a mean of 2.987 at the final checkpoint, versus 7.519 under nf4—a 2.52$\times$ backend ratio. The logged `val_margin` values are therefore not comparable to the fp16 held-out margins in Table 13.